1. $\ell = -\left[y \log h(\theta^T x) + (1-y) \log (1 - h(\theta^T x)) \right]$

$$\frac{\partial \ell}{\partial \theta_j} = \frac{d\ell}{d h(\theta^T x)} \cdot \frac{d h(\theta^T x)}{d \theta^T x} \cdot \frac{\partial \theta^T x}{\partial \theta_j}$$

← Chain rule (θ changes ℓ through h & and z through $h(\theta^T x)$)

$$\frac{d\ell}{d h(\theta^T x)} = -\left[y \log h(\theta^T x) + (1-y) \log (1 - h(\theta^T x)) \right]$$

$\frac{d \log x}{dx} = \frac{1}{x}$; chain rule

$$= -\left[y \cdot \frac{1}{h(\theta^T x)} + (1-y) \cdot \frac{1}{1 - h(\theta^T x)} (-1) \right]$$

$$= -\frac{y}{h(\theta^T x)} + \frac{1-y}{1-(h(\theta^T x)}$$

$$= \frac{-y + h(\theta^T x)}{h(\theta^T x)(1 - h(\theta^T x)}$$

$$\frac{-y}{h} + \frac{1-y}{1-h} = \frac{-y(1-h) + h(1-h)}{h(1-h)} = \frac{-y + yh - yh + h}{h(1-h)} = \frac{-y+h}{h(1-h)}$$

Sigmoid (chain rule)

$$\frac{d\, h(\theta^T x)}{d \theta^T x} \quad \frac{1}{1 + e^{-\theta^T x}} = (1 + e^{-\theta^T x})^{-1}$$

$$= -(1 + e^{-\theta^T x})^{-2} \cdot (-e^{-\theta^T x})$$

$$= \frac{e^{-\theta^T x}}{(1 + e^{-\theta^T x})^2} = \frac{1}{1 + e^{-\theta^T x}} \cdot \frac{e^{-\theta^T x}}{1 + e^{-\theta^T x}}$$

$h(\theta^T x)$; $1 - h(\theta^T x)$

$$= h(\theta^T x)(1 - h(\theta^T x))$$

$$\frac{\partial \theta^T x}{\partial \theta_j} = \frac{\partial \theta^T x}{\partial \theta_0} \cdot \frac{\partial \theta^T x}{\partial \theta_1} \cdot \frac{\partial \theta^T x}{\partial \theta_2} \rightarrow \theta^T x = \begin{bmatrix} \theta_0 \\ \theta_1 x_1 \\ \theta_2 x_2 \end{bmatrix}$$

$$\frac{\partial \theta^T x}{\partial \theta_0} = 1, \quad \frac{\partial \theta^T x}{\partial \theta_1} = x_1, \quad \frac{\partial \theta^T x}{\partial \theta_2} = x_2 \quad = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$

$$\frac{\partial \ell}{\partial \theta_j} = \frac{-y + h(\theta^T x)}{h(\theta^T x)(1 - h(\theta^T x)} \cdot h(\theta^T x)(1 - h(\theta^T x)) \cdot \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$

$$= -y + h(\theta^T x) \cdot \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$

$$\nabla_\theta \ell(\theta) = (-y + h(\theta^T x)) \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix}$$